Steam Tag Explorer
Data Visualization Project Report

Michal Kubirita Marina Ramírez Baños Matej Zelenák

December 12, 2025

Abstract

The rapid expansion of Steam's game catalogue and user base has generated large amounts of publicly accessible data, offering valuable opportunities for analysing trends in player engagement and market saturation. In this project, we present an interactive visualization tool designed to help game developers and non-expert users explore which Steam game tags attract the most user attention and how competitive each tag's market is over time. Using a dataset of over 111,000 games and 453 unique tags, our tool integrates multiple metrics such as peak concurrent users, review statistics, release counts, pricing, and playtime. Through coordinated visualizations, filtering operations, and detail on-demand interactions, users can compare tags, identify temporal patterns, and investigate relationships between key variables. The resulting system provides an accessible and effective data-driven approach for understanding trends in the Steam marketplace.

1 Introduction and Motivation

Since the last decade the video game industry has experienced rapid growth. Steam [1], owned by Valve Corporation [2], is currently the most used platform for digital games distribution on personal computers. Thousands of games of all types are released every year on the platform, having published more than 8000 games only in 2018 [3].

As Steam's catalogue continues to grow, so does its user base, generating large amounts of data. In an expanding and competitive market like video game development, data-driven analysis of user behaviour and game popularity is essential for developers and publishers to understand their customer's needs [4].

In this project we developed an innovative, user-friendly visualization tool[1] that aims to help game developers determine which type of game has the greatest potential to capture Steam consumer's interest. Specifically, our main goal is to help the user identify which game tags on the platform tend to capture user's attention, and which type of games, based on these tags, face a less saturated market, based on annual game releases.

Our visualization tool uses a Steam dataset obtained from Kaggle [5], which is created using the official Steam API. The dataset contains 111,452 games published from 1997 to 2025, 453 unique game tags, and different metrics including the games' release dates, range of estimated number of owners, peak of concurrent players (CCU), price, and average playtime.

2 Visualization Problem

The main goal of our visualization was to enable users to answer two research questions: "Which Steam game tags tend to capture users' attention?" and "How saturated is the market for those tags?". Based on this, we defined the data context and data content of our visualization.

Our data context included temporal data, such as the year of release of the games (from 1997 to 2025), and categorical data, such as the game tags (453 tags). The year of release provides users a temporal frame to explore the data, while the game tags are associated with multiple parameters users

[1] `https://steamtaganalysis.streamlit.app/`

can investigate. Game tags data exhibit a network topology, since each game can be associated with multiple tags.

Among the various variables included in the dataset, we identified the following as our data content to achieve the stated goal of the visualization (see Table 1).

Metric	Data Type	Description
Peak CCU	Quantitative discrete, scalar	Maximum number of concurrent players in a game
Total reviews / positive reviews	Quantitative discrete, scalar	Indicates users' perception of the game tag
Average playtime (minutes)	Quantitative continuous, scalar	Indicates how much time users spent playing games within a tag
Range of estimated owners	Ordinal, vector	Estimated ownership range for games in a tag
Number of released games	Quantitative discrete, scalar	Indicates how saturated the market is within a game tag
Total Free-to-Play games	Quantitative discrete, scalar	Additional detail about the tag's market
Price	Quantitative continuous, scalar	Price information of games in the tag
Game names	Qualitative nominal, scalar	Names of games within a tag

Table 1: Summary of metrics for analysing Steam game tags

Our visualization needed to be implemented so that users could get an overview of the data, zoom in and filter to further explore a selected game tag, and obtain details on demand about that tag. Therefore, users should be able to perform the following main tasks:

Task 1: Elementary identification of a game tag. Identify where a game tag lies in the data context. Specifically, find a game tag that has a high peak CCU and faces a less saturated market in terms of the number of games released within a given time period.

Task 2: Elementary comparison of game tags. Compare a subset of selected game tags based on different variables, including peak CCU, average positive reviews, and number of released games.

Task 3: Synoptic pattern comparison. For a selected game tag, compare the distribution of different variables over time - for instance, distribution peak CCU and price over time.

The visualization is primarily intended for use by game developers for market analysis, but we also wanted to allow the general public to explore the data. Therefore, the visualization includes clear, non-technical descriptions to guide the user where necessary.

3 Related Work

There are multiple existing works visualizing video games on the Steam platform [6; 7; 8; 9]. However, we have found limitations in all of these tools, such as low or no interactivity, missing a clear goal with a visualization question in mind, limited view of tags and video games preselected by the authors. That is why we have chosen to make our own tool aiming to avoid all of the above limitations while having a clear visualization goal and purpose.

In our visualization tool, we have used an UpSet chart, which is a visualization of intersecting sets created at Harvard in 2014 [10]. It is a more readable alternative to the Venn diagram because it is better at showing more than three sets that intersect, which is a common pitfall of Venn diagrams.

4 Interactive Visualization Solution

4.1 Operations

The main visualization (Main Overview) interface is divided in 4 different panels: (A) left panel with filters, (B) dataset summary, (C) scatter plot "Peak CCU vs Number of Released Games per Tag", (D) list containing all the tags' names included on the original dataset. The main visualization provides the user the possibility of adjusting the visualization from viewing to exploring and obtain details on demand by performing the following operations:

- **Filter on the left panel.** The left panel includes three sliders that the user can adjust by clicking and dragging to filter the information displayed on the panels B and C. The user can adjust the time interval shown using the "Year Range" slider, as well as the amount of game tags shown using the "Minimum amount of reviews per game" and "Minimum amount of peak CCU per game" sliders.
- **Write and select.** The user has the possibility for issuing visual queries, specified game tags, typing the name on the search bar under the description "Tags to highlight" on panel C. By typing the user can select one or multiple game tags that will be then highlighted in red on the scatter plot below.
- **Click and select.** The user can click on a specific point on the scatter plot of panel C, which corresponds to a game tag, and go to the "Tag Details" tab, to obtain details on demand.
- **Hover.** A small window containing details about a game tag is displayed when hovering over a point on the scatter plot of the panel C.
- **Zoom in.** It is possible to zoom in on the scatter plot of panel C by clicking on the "+" icon or by using the lens, clicking the magnifying glass icon, then selecting and dragging over the plot.

The secondary visualization (Tag Details) provides details about a selected game tag on the scatter plot of panel C. The interface is six panels: (E) summary of relevant metrics; (F) bar plot "Number of Games Released Over Time"; (G) box and whisker plots of metrics over years; (H) most common categories and most common tags associated with the selected tag tables; (I) UpSet [10] plot showing tag intersections of the selected tag based on the number of games; (J) table containing all games included on the selected tag. The user can adjust the visualization by performing the following operations:

- **Filter on the left panel.** Similarly to the main visualization, the left panel (A) can also be used to filter the data displayed in all panels at once.
- **Hover.** On panels F and G it is possible to obtain more details when hovering over a bar.
- **Zoom in.** On panels F and G, it is possible to zoom in in the same way described for panel C.
- **Write and select.** Panel I includes a search bar that is predefined to select the top 5 most commonly tags associated with the selected tag and the selected tag. It is possible to remove and add tags by writing and selecting on the displayed list of options.

4.2 Visual Encoding

In this section, we describe the visual encoding used in our visualization tool and justify our design decisions.

4.2.1 Main Visualization (Main Overview)

The main visualization includes two panels, B and D, that we decided not to plot, as they only provide additional contextual information about the original dataset and the displayed data after filtering.

Scatter Plot - Peak CCU vs. Number of Released Games per Tag. Each game tag is represented as a point. The vertical position encodes the average peak CCU, and the horizontal position encodes the number of released games. The axes are log-scaled due to the wide dispersion of the data.

We chose a scatter plot because it allows users to easily recognize which game tags have a higher number of concurrent players and face a less saturated market (top-left area of the panel), supporting Task 1. Additionally, points (game tags) can be compared across plots based on their position, supporting Task 2. Selected game tags are highlighted in red to facilitate identification, as red is an attention-grabbing colour that contrasts strongly with the light blue used for unselected tags.

4.2.2 Secondary Visualization (Tag Details)

Similarly to the main visualization, panels E, H, and J do not contain plots. Panel E provides contextual information about the original data and the displayed subset; panel H summarizes associated tags and categories; and panel J lists all individual games in the selected tag, which is not the focus of the visualization tool.

Bar Plot - Number of Games Released Over Time. Panel F contains a bar plot showing the number of games released per year within the selected tag. Because the data are grouped by year, the time axis is treated as a discrete set of points.

We used a bar plot to highlight the number of games released each year, enabling straightforward comparison of bar lengths. This helps users identify periods of higher market saturation for the selected tag, and on which years most of the games included were released.

Box and Whisker Plots - Metrics Over Years. Panel G shows box plots of average playtime, average ratio of positive reviews, peak CCU, and price plotted against the release years for the selected tag. The goal is to help users compare patterns between those variables, supporting Task 3.

We "linked" the plots, automatically adjusting the zoom to facilitate exploration. Box plots were chosen because they clearly summarize data distribution (median, Q1, Q3, minimum, maximum).

UpSet Plot - Tag Intersections with the Selected Tag. Panel I presents an UpSet [10] plot displaying tag intersections and number of released games. Each vertical bar indicates the size of an intersection (i.e., the number of games shared across specific tag combinations), sorted from largest to smallest. Tags participating in each intersection are shown using black points beneath the bars. The number of games per tag is represented with horizontal bar plots next to the tag name.

An UpSet plot was chosen because it effectively shows overlaps among selected tags and is currently one of the best methods for visualizing intersections of qualitative categories. Its use of sorted vertical bars highlights the largest intersections, and its ability to incorporate multiple categories enhances exploration.

4.3 Interaction

We now describe the different interaction strategies used and how they support the tasks mentioned, allowing users to analyse and explore the data:

Reversible filters that can be layered for a tailored visualization. The filters of panel C affect all the information displayed in the panels, and can be added one on top of each other. That allows users to obtain a visualization tailored to the needs of the analysis to be performed.

Writing to selection and plot highlighting. The search bar used on panel C interacts with the scatter plot highlighting the selected game tags. This supports Task 1 and Task 2, as it visually allows the user to identify and compare multiple game tags based on the position of the highlighted points of the graph.

Click and select to open details on demand. On an intuitive and simple approach, users can go from an overview visualization to a details-on-demand visualization by clicking on a point of the scatter plot of panel C (game tag).

Ability to move between overview and details-on-demand visualizations. On panel C, users can select the visualization they want to display, allowing them to rapidly switch between visualizations.

Connected interactive graphs. On panel G, users can observe and compare the distribution of different parameters for the same games within a tag and released in the same year. All box and whisker plots are automatically scaled when zooming in into one of the plots, allowing the user to perform a side to side comparison, supporting Task 3.

4.4 Implementation Challenges

Throughout development, several practical challenges emerged that influenced both the design and performance of our visualization tool. A primary difficulty was the size of the dataset: the original CSV contains more than 111,000 games and 453 tags. Processing this data on the fly, especially generating aggregated statistics per tag and per year, proved computationally expensive. To maintain a responsive interface, we implemented pre-aggregation steps (e.g., computing tag-level summaries in advance) and optimized filtering operations to avoid recomputing expensive group-by operations after every slider interaction.

Rendering performance also required attention. The main scatter plot can display hundreds of data points, while the Tag Details tab may draw dozens of box plots and bar plots simultaneously. When rendered naïvely, these visualizations introduced noticeable delays during while choosing large tag groups. We addressed this by reducing unnecessary redraws, memorizing intermediate results, and relying on efficient rendering strategies provided by our visualization libraries.

Another significant challenge involved the UpSet plot. Standard plotting libraries do not natively support this visualization, so tag intersections and visual mappings had to be implemented manually. Because many tags frequently co-occur, computing all possible intersections results in a combinatorial explosion.

Finally, coordinating multiple interacting views introduced additional complexity. Filters and tag selections needed to update all relevant panels consistently. Ensuring that these linked interactions behaved predictably required careful state management and several iterations of debugging, refinement and optimisation.

5 Example Use Case

During the whole design process, we had a use case in mind where a game developer wants to do a business/market analysis to know if their next video game will have potential on the Steam platform. We describe an example scenario below.

1. The user is interested in finding out whether multiplayer games receive more attention and positive engagement than singleplayer ones.

2. In the main overview's scatter plot chart, they can see that online co-op games have a higher ratio of positive reviews opposed to multiplayer or singleplayer games. In addition, less games are labelled with online co-op than the other two. For a game developer, this is good because they might have an easier chance of standing out with less competition on the market.

3. Clicking on the tag, the user is sent into the tag details page, where they can find out that games released in the last four years have higher peaks of concurrently playing players than in the past, although the playtime has been decreasing. This shows that there has recently been an audience for the selected tag.

4. In the common categories table, the user can find out that games with full controller support have twice as high peak of concurrent players than just with partial controller support.

5. In the UpSet plot, the user can identify the competition in more detail by choosing additional tags. For example, there are almost twice as many shooter online co-op games than fantasy online co-op ones.

6. Lastly, if the user is curious about specific games affecting the data, they can look them up in the table of all games of the selected tag. For example, they could see in the box and whisker plots of peak concurrent players that year 2006 and 2007 are particularly strong and together consist of only 5 games. To find out what is behind this, the user searches for them in the games table and finds: Team Fortress 2; Garry's Mod; Call of Duty® 2; DOOM II; Company of Heroes: Opposing Fronts. Afterwards, they could conduct their own further research into why did these become popular.

6 Discussion

6.1 Limitations of the Data

The measure of users' attention to a specific game tag is not based on a unique, widely used parameter. We had to decide and select a group of variables that could be informative about the users' perception and interaction with the games. From that subset, we consider that the peak CCU was the most insightful variable, as it provides an idea of the maximum number of users that actually interacted/played the games. Other variables, such as the average number of positive reviews and the average playtime forever, does not directly indicate how many users played the game. We did not use the estimated number of owners because of the type of variable provided: it consisted of wide ranges, which would have made it difficult to get insightful information.

6.2 Limitations of the Visualization Design

To evaluate the limitations of the visualization design we perform a user study case. We provided an online form that requested users to assess the visual layout (colour palette, UI, etc.), comprehensibility, interactiveness, and usability of the visualization. The user was also requested to perform a small analysis and describe the findings. The form was completed by nine users whose technical knowledge in game development or data visualization ranged from none to basic.

All users were able to understand the purpose of the visualization tool and perform a data analysis. Users acknowledge the comprehensiveness and usability of the visualization for the general public (score: 8.56/10) and game developers (score: 8.78/10) to explore Steam market, and potentially make decisions about their next games. Regarding task abstraction, users were able to perform all required operations, and gave a high score when asked about the interactiveness of the visualization (score: 9/10). However, a minor limitation was addressed: "when changing the filters of the left panel, tag selection resets to none".

The visual encoding was qualified as adequate, clean, and simple. Nevertheless, some limitations were mentioned: one user indicated that "it would be helpful to have different highlight colours on the main visualization when selecting multiple tasks". Also, the information displayed about different metrics on the "Tag Details" tab "looked a bit overwhelming".

Regarding performance, one user experienced a slight delay when loading the main visualization. This issue is primarily caused by the need to load a large portion of the dataset and generate the plot on the fly.

7 Contribution of Group Members

A major part of the design decision process was done together. Table 2 shows individual contributions of each group member.

Group Member	Contribution	Design Choice
Michal	Research of related work	Box and whisker plots
Marina	Layout design	Scatter plot
Matej	Coding, data pre-processing	UpSet plot

Table 2: Individual contributions and design choices within the projet group.

References

[1] Valve Corporation. Steam. `https://store.steampowered.com/`, 2025. Accessed: 11 December 2025.

[2] Valve Corporation. Valve corporation. `https://www.valvesoftware.com/`, 2025. Accessed: 11 December 2025.

[3] Andraž De Luisa, Jan Hartman, David Nabergoj, Samo Pahor, Marko Rus, Bozhidar Stevanoski, Jure Demšar, and Erik Štrumbelj. Predicting the popularity of games on steam. *arXiv preprint arXiv:2110.02896*, 2021.

[4] Eric J Toy, Jaya VHH Kummaragunta, and Jin Soung Yoo. Large-scale cross-country analysis of steam popularity. In *2018 International Conference on Computational Science and Computational Intelligence (CSCI)*, pages 1054–1058. IEEE, 2018.

[5] fronkongames. Steam games dataset. `https://www.kaggle.com/datasets/fronkongames/steam-games-dataset`, 2025. Accessed: 11 December 2025.

[6] itai-reder. Steam games data analysis project. `https://steam-games-visualization.streamlit.app/`. Accessed: 2025-12-11.

[7] Alexander Kainz and Johanna Pirker. Steamvis: a tool for collecting and analyzing data of the game distribution platform steam. In *2023 IEEE Conference on Games (CoG)*, pages 1–8. IEEE, 2023.

[8] aaw707. Steam games visualization. `https://github.com/aaw707/Steam-Games-Visualization`. GitHub repository, accessed: 2025-12-11.

[9] Claro Henrique and Natã Raulino. Steam visualization: Perspectives on what makes a game popular. `https://clarohenrique.github.io/steam_data_visualization/`. Interactive Steam games data dashboard, accessed: 2025-12-11.

[10] Alexander Lex, Nils Gehlenborg, Hendrik Strobelt, Romain Vuillemot, and Hanspeter Pfister. Upset: Visualization of intersecting sets. *IEEE Transactions on Visualization and Computer Graphics (InfoVis)*, 20(12):1983–1992, 2014.